MouseLFP: current network architecture

Fixed spectral basis ⟶ shared residual frontend ⟶ task-local CNN. Shapes include batch size B.

1 Fixed preprocessing (no fitted normalization statistics)

LFP window
112 s at 500 Hz
$B \times 56{,}000$

→

Center → resample → STFT
Per-window mean removal; 500 → 256 Hz
Cosine window 256; hop 128; end zero-padding
Unnormalized magnitude; retain 1–40 Hz

→

Magnitude M
$B \times 40 \times 223$
Fixed basis cached

→

Fixed log transform + transpose
$L = \log(1 + M)/\ln(10)$
Time × frequency map: $B \times 1 \times 223 \times 40$

2 Shared convolution + MLP residual frontend F_θ (49,704 learnable parameters)

Conv2d + GELU
$1 \to 16$, 3×3
Stride $(1, 1)$, pad 1
$B \times 16 \times 223 \times 40$

→

Conv2d + GELU
$16 \to 16$, 3×3
Stride $(2, 1)$, pad 1
$B \times 16 \times 112 \times 40$

→

Conv2d + GELU
$16 \to 16$, 3×3
Stride $(2, 1)$, pad 1
$B \times 16 \times 56 \times 40$

Sample-local features; frontend has no BatchNorm or dropout

→

Per-time-bin MLP
Reshape to $B \times 56 \times 640$; LayerNorm(640)
Linear $640 \to 64 \to$ GELU $\to$ Linear $64 \to 40$
Last layer: zero weights and bias
Residual: $B \times 1 \times 56 \times 40$

$F_\theta(M)$ ⊕

Residual base: adaptive average pool L to $B \times 1 \times 56 \times 40$
Frequency axis stays at 40 bins; absolute amplitude is retained

CNN input: $B \times 1 \times 1 \times 56 \times 40$ (channel, electrode, time, frequency)

3 Task-local CNN C_{w_m}: reset from a fixed unlearned reference for each mouse

CNN block 1
AxisBN (electrode axis)
Conv3d $1 \to 16$, $1 \times 5 \times 5$
Stride $(1, 2, 2)$, pad $(0, 2, 2)$
ReLU; max pool $1 \times 2 \times 2$
Ceil mode; squeeze E
$B \times 16 \times 14 \times 10$

→

CNN block 2
AxisBN (16 channels)
Conv2d $16 \to 32$, 3×3
Stride 1, padding 0
ReLU; max pool 2×2
$B \times 32 \times 6 \times 4$

→

CNN block 3
AxisBN (32 channels)
Conv2d $32 \to 64$, 3×3
Stride 1, padding 0
ReLU; max pool 2×2
$B \times 64 \times 2 \times 1$

→

Dense classifier
Flatten $\to 128$ features
Dropout $(p = 0)$
Linear $128 \to 128$
Sigmoid
Dropout $(p = 0)$

→

Binary prediction
Linear $128 \to 2$: logits $B \times 2$
Training: cross-entropy on logits
Inference: softmax $\to p(y = 1)$
Accuracy threshold: 0.5

Initialization: $F_{\theta_0}(M) = \mathrm{AvgPool}_{56\times40}(L)$, exactly the pure log-STFT baseline; initialization preserves caller RNG streams.

CNN state: support uses training-mode BatchNorm; query/test use eval mode with support-fitted statistics (EMA retention 0.99, $\epsilon = 10^{-3}$). All CNN fitting uses dropout $p = 0$.

Implementation: `meta_learning/frontend.py`, `meta_learning/model.py`, `models/cnn.py`. Teal: shared frontend parameters θ; blue: mouse-specific CNN parameters w_m. Snapshot: 2026-10-16.

MouseLFP: frontend-only bilevel learning

One fixed task per configured mouse; support fits the CNN, query supervises the frontend, and test is inference only.

Fixed temporal split for each mouse m; all eligible support/query windows in the current configuration

Support S_m: Days 0–7
CNN fitting and support-only BatchNorm statistics

Validation query Q_m: Days 8/11
Outer frontend gradient and checkpoint selection

Test eval E_m: Days 12/13
Inference and reporting only

One task per mouse, once in every outer step

Shared frontend $F_\theta(S_m)$
Fixed θ during this task's inner fitting

Same frontend $F_\theta(Q_m)$
Same θ for every mouse in this outer step

Fit task-local CNN on support labels
$w_m^{(K)}, b_m^{(K)} = \mathrm{Fit}_{50\ \mathrm{epochs}}(w_0, b_0; F_\theta(S_m), y_{S_m})$
SGD lr 0.01, momentum 0.9; full shuffled epochs
Keep weights, momentum, and BN state within task
K: all minibatch updates in the 50 epochs

fitted CNN + BN

Query forward in CNN eval mode
$\ell_m(\theta) = \mathrm{CE}_{0.1}(C_{w_m^{(K)}, b_m^{(K)}}(F_\theta(Q_m)), y_{Q_m})$
Support-fitted BN statistics; dropout disabled
Query labels supervise θ through the full fit

Fixed unlearned reference (w_0, b_0)
Reset for each mouse / outer step
CNN initialization is never meta-learned

Full frontend hypergradient $g_m = \nabla_\theta \ell_m$
Direct query path + all support-training updates
Includes support BatchNorm EMA trajectory
RNG-preserving replay: 5 updates per segment

next outer step

One outer update after all configured mice
$g = \frac{1}{M}\sum_{m=1}^{M} g_m, \quad \theta \leftarrow \mathrm{Adam}(\theta, g; \eta = 10^{-4})$
Equal mouse weights; gradient norm clipping at 5; only frontend parameters update

Every 10 outer steps and at the final step

Fresh support-only fit for every mouse
Freeze the updated shared frontend θ
Reset CNN to (w_0, b_0); fit on Days 0–7
Same optimizer and 50-epoch budget

Evaluate both query and test
Days 8/11: query log loss and accuracy
Days 12/13: test log loss and accuracy
Eval mode, frozen BN statistics, no gradients
Accuracy uses probability threshold 0.5

query criterion

Select by mean query log loss
Minimum equally weighted mouse mean
After the matched support-only refit
Ordinary unsmoothed predictive log loss
Save shared frontend + fitted per-mouse CNNs

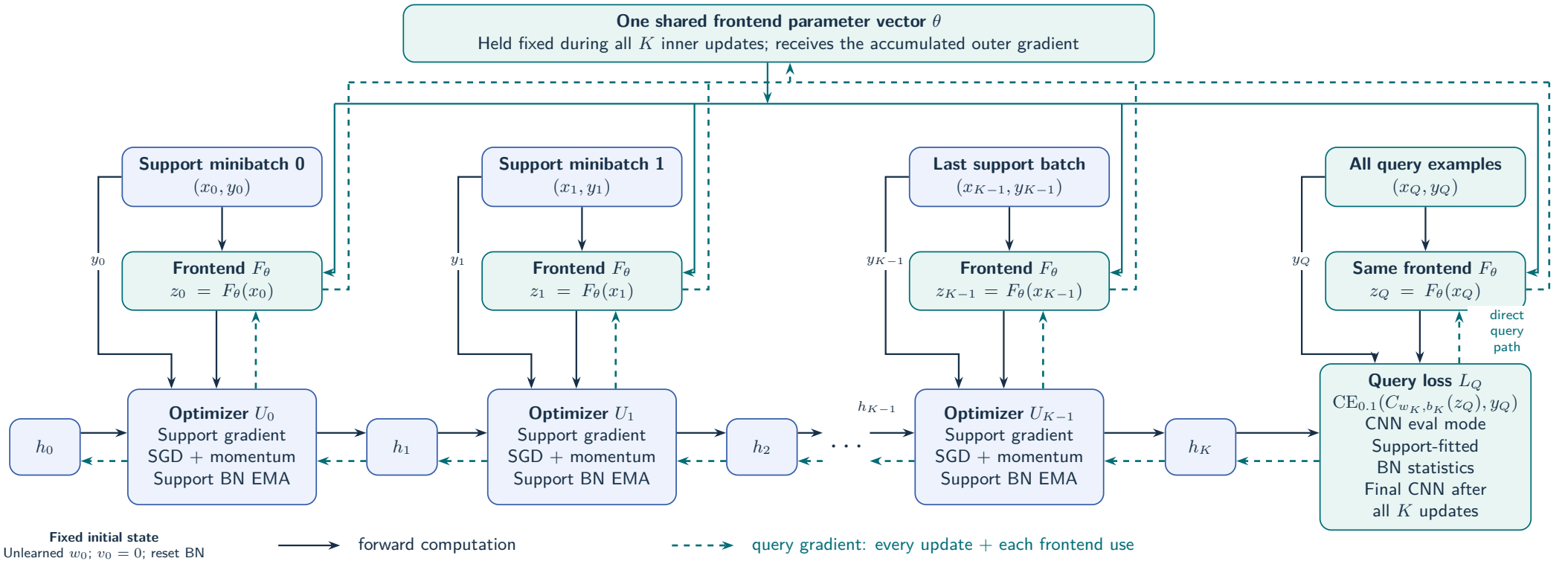
Independent control: `log_stft`
Parameter-free pooled log-STFT; fit one CNN per mouse once.
Query/test inference after fitting; no outer updates or startup baseline.

Matched control: `pooled`
Same frontend/CNN; direct supervision on support + query labels.
Same mouse schedule; discard auxiliary CNN before fresh support-only refitting.

Current `configs/meta.json`: targets=[X2], $M = 1$, 1,000 outer steps, batch size 32, 50 inner epochs, fixed shuffle seed 0, label smoothing 0.1. Full cohort: X1/X2/X4/X9/X10/X14/X15 ($M = 7$).

MouseLFP: inner optimization unrolled as a recurrent graph

One mouse, one outer step. The recurrent state is the CNN, momentum, and BatchNorm state; each transition consumes one support minibatch.



The recurrent state and update rule

$h_t = (w_t, v_t, b_t)$: CNN weights, momentum, BN statistics

$$\widetilde{g}_t = \mathrm{ClipNorm}_5\left(\nabla_{w_t} \mathrm{CE}_{0.1}(C^{\mathrm{train}}_{w_t}(z_t), y_t)\right)$$

$$v_{t+1} = 0.9 v_t + \widetilde{g}_t, \qquad w_{t+1} = w_t - 0.01 v_{t+1}$$

$$b_{t+1} = 0.99 b_t + 0.01\,\widehat{b}_t \quad \text{(support batch means / biased variances)}$$

$K = 50\,\lceil |S_m|/32 \rceil$: all minibatches of all 50 complete epochs

Backpropagation through the complete optimizer trajectory

$$\lambda_K = \nabla_{h_K} L_Q, \qquad \lambda_t = \left(\frac{\partial U_t}{\partial h_t}\right)^\top \lambda_{t+1}$$

$$\nabla_\theta L_Q = \underbrace{\nabla_\theta L_Q|_{h_K \text{ fixed}}}_{\text{direct query frontend path}} + \underbrace{\sum_{t=0}^{K-1} \left(\frac{\partial U_t}{\partial \theta}\right)^\top \lambda_{t+1}}_{\text{all support-training paths}}$$

$$h_{t+1} = U_t(h_t; \theta) = U(h_t, F_\theta(x_t), y_t).$$

Replay recomputes 5-update segments and preserves the full gradient.
Outer optimization updates only θ; each new task resets h_0.

Support: Days 0–7. Query: Days 8/11. All intermediate updates represented by the ellipsis participate in backpropagation. Test Days 12/13 are inference only. Implementation: `meta_learning/inner.py` and `meta_learning/algorithms.py`.